Deepak Mallampati

GENERATIVE AI & RAG ENGINEER

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | GitHub |

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building production Generative AI systems - RAG over large document corpora (dense retrieval + cross-encoder re-ranking), agentic multi-agent workflows, and LLM evaluation across AWS, Azure, and Databricks. Delivered measurable outcomes including 42% token reduction, 40% hallucination reduction, and 92% retrieval precision. Strong fit for document-intelligence, knowledge-retrieval, content-generation, and workflow-automation engineering.

CORE COMPETENCIES & SKILLS

RAG Architecture • Vector Databases (FAISS/Pinecone/Weaviate) • Agentic Workflows • LangChain / LangGraph • MCP • Embeddings & Semantic Search • LLM Evaluation • Workflow Automation

Generative AI & RAG: LLMs (GPT-4o, Claude, Gemini, LLaMA), RAG architecture, embeddings, cross-encoder re-ranking, semantic & hybrid search, LangChain/LangGraph, MCP, multi-agent systems, Hugging Face Transformers

Vector & Retrieval: FAISS, Pinecone, Weaviate, document chunking, semantic caching

Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI/AI Services, GCP Vertex AI, Databricks, Docker, Kubernetes, MLflow, Weights & Biases

ML & Evaluation: PyTorch, TensorFlow, RAGAS, BERTScore, drift detection, A/B testing

Languages: Python, JavaScript/TypeScript, SQL, FastAPI

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard *Jan 2024 - Jan 2025*

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%**.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University *Oct 2023 - Jan 2024*

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical readmission dataset (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset for controllable long-form generation.

Emerson *Jun 2022 - Apr 2023*

ML Trainee

India

- Developed supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting on operational pipeline-storage data.
- Engineered features and end-to-end preprocessing pipelines; fine-tuned **BERT and RoBERTa** for NER and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - compressing hours of repetitive work into an automated workflow.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across 29+ sites, turning a multi-step manual chore into a single inline action.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy that quantifies exactly how much privacy each technique buys for how little accuracy cost.

EDUCATION

M.S. in Computer Science - *Kent State University, OH* *2025*

GPA: 3.70 / 4.0